

Demography and social learning impact norm misperception and belief revision following intervention

Study materials and protocol

This companion compiles the documented measures, source framing, and procedures relevant to the released data. Question wording is reproduced from the source records where available; procedural descriptions summarize the manuscript and supplement. It is not an executable Qualtrics project or a reconstruction of unprovided screen layouts and survey logic. Sources are identified in Section 7.

1. Study overview

The main study was a two-stage online experiment with 903 U.S. adults and five conditions: Individual (no-feedback control), Age, Conformity, Prestige, and Season. Participants first reported their own agreement with 12 gender-norm statements and estimated others' agreement. They then reviewed their estimates, with or without source-attributed reference values, and could retain or revise them. The reference values came from a separate survey sample of 879 participants. [1, 2, 4]

1.1 Main-study sequence

After study information and informed consent, participants supplied demographic information, indicated personal agreement with the 12 statements, and estimated the percentage of the specified reference population agreeing with each statement. They then rated their overall confidence in those initial estimates. [1, 2]

The manuscript reports random assignment, stratified by gender, age, and education, to one of the five conditions. The four feedback conditions received the same statement-specific numbers under different source labels. The control condition received no reference values. Participants reviewed their initial estimates and could revise them. The subsequent measures included an open-ended explanation and four auxiliary social-influence items; the manuscript reports debriefing before exit. [1–3]

1.2 Reference population and incentives

The estimation task described a population of 1,000 adults living in the United States, aged 18–64 and balanced by gender, age group, and education. This is the population participants were asked to estimate, not the attributed source subgroup. Estimates used a 0–100 percentage scale. The supplement reports a \$3 participation payment and an additional \$10 for each of the ten most accurate participants. [1, 2]

The released main-study files start with the cleaned analytic sample. The statement identifiers in this document match the release files; they do not indicate the order in which statements were presented.

2. Statement bank

Six statements express equitable norms and six express inequitable norms. Statement texts and reference values below reproduce data/statements.csv and Supplementary Table S1. Agreement refers to Agree or Strongly agree on the five-point scale. References are rounded pilot agreement percentages and are identical across the four feedback conditions. They are not separate subgroup estimates. [1, 2, 4]

Table 1. Gender-norm statements and displayed references.

ID	Statement	Reference (%)
1	A woman should be free to refuse sex with her husband or partner under any circumstances.	93
2	It is important for girls and women to be educated.	96
3	A woman should be free to divorce or leave her husband even if he does not wish to end the marriage.	91
4	Men and women are equally suitable for all professions.	60
5	It is important for women to have more opportunities in politics.	71
6	A woman can live a successful life even if she does not marry a man.	95
7	Women often exaggerate their accounts of sexual violence in order to take advantage of the situation.	10
8	A university education is more important for a boy than for a girl.	5
9	Even if a mother is being mistreated by her partner, she should stay for the sake of the children.	3
10	When women lose out to men in a fair competition, they like to claim that they have been discriminated against.	15
11	Men make better political leaders than women do.	15
12	A woman has to have children in order to be fulfilled.	8

Statements 1–6 are equitable as worded; statements 7–12 are inequitable. Domain pairs are sexual autonomy (1, 7), education (2, 8), family autonomy (3, 9), professional equality (4, 10), political participation (5, 11), and marital autonomy (6, 12). [2]

3. Main-study questionnaire measures

3.1 Demographic prompts

The following prompts are copied from the survey export's question-label row. The release field name is provided to identify the corresponding cleaned variable. Response coding and observed category labels are documented in DATA_DICTIONARY.md; the export does not supply a complete unselected answer-option list. [3]

Table 2. Demographic question wording.

Release field	Question
Gender	Are you ____?
Age	What is your age? (Please enter your age as a whole number in years — for example, enter 21 if you are 21 years old)
Education	What was the highest level of education that you have attended?
Marital Status	Which of the following best corresponds to your current situation?
Race	What is your race or ethnicity? (Please select all that apply)

Release field	Question
Political Affiliation	Which of the following best describes your political affiliation?
Favorite Season	What is your favorite season?

3.2 Personal agreement

Participants reported their personal agreement with each statement using the response categories Strongly disagree, Disagree, Neither agree nor disagree, Agree, and Strongly agree. These are coded 1 through 5 in the release. The six equitable items are not reversed in the stored responses; reverse coding is performed only where required for the common gender-inequity outcome. [1–3]

3.3 Initial and final estimates

The exported main-study slider label is: [3]

Estimated agreement (What % of other study participants do you think would agree with the above statement?)

The broader target-population description is given in Section 1.2. Initial and final estimates refer to agreement with each statement as worded. A final estimate may equal the initial estimate. The manuscript describes revision as optional, rather than a requirement to change the numerical answer. [1]

3.4 Confidence in initial estimates

One overall confidence response was collected after the initial-estimate task. The question wording in the export is: [2, 3]

Lastly, how confident are you about your estimates regarding other participants' responses?

Please answer based on your overall impression across the full set of statements.

The recorded categories are Not at all confident, A little confident, Somewhat confident, Quite confident, and Very confident. This auxiliary measure is stored in data/confidence.csv and is not used by the six main-analysis scripts.

3.5 Open-ended explanation

The exported question wording is: [3]

Why did you choose to revise or keep your original estimate in the previous section?

(You may write just a sentence or two)

The response texts and coding records are stored in data/qualitative_coding_final.xlsx. The coding-stage distinction and frequency procedure are specified in Section 6.

3.6 Auxiliary social-influence items

The supplementary procedure diagram places these items after the revision task. They concern product-related information seeking and are stored in data/ssi.csv as Q1–Q4. The wording below is copied from the export. [2, 3]

Table 3. Auxiliary social-influence items.

Release item	Statement
Q1	I often consult other people to help choose the best alternative available from a product class.
Q2	To make sure I buy the right product or brand, I often observe what others are buying and using.
Q3	If I have little experience with a product, I often ask my friends about the product.
Q4	I frequently gather information from friends or family about a product before I buy.

The recorded response scale is Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, and Strongly agree. Q3/Q4 store the last label as “Strong agree.” The saved composite is the mean after mapping the five categories to 0, 0.25, 0.50, 0.75, and 1. All 902 released records have four responses. These items are shared as auxiliary measures, not as outcomes in the six main-analysis scripts.

4. Social-information conditions

The numeric reference is drawn from the pilot agreement data, but participants encounter it as an estimate attributed to a source. The source label varies by condition; the population being estimated and the reference number for a given statement do not. The source phrases below are copied from the exported feedback wording. Control is described from the manuscript. [1–3]

Table 4. Source-attribution wording. [reference] denotes the statement-specific value in Table 1.

Condition	Information
Individual (control)	No source-attributed reference value is shown. Participants review their initial estimates and may revise them.
Age	Here is the average estimate provided by participants aged 40 and above: [reference]%
Conformity	Here is the most common estimate provided by participants in this study: [reference]%
Prestige	Here is the average estimate provided by participants with a graduate or professional degree: [reference]%
Season	Here is the average estimate provided by participants who like summer: [reference]%

4.1 Example of exported feedback wording

The following Conformity example is copied from the export for the political-opportunity statement. The bracketed Field-E7_Response token is the saved dynamic-text placeholder for the participant’s previous estimate, not a displayed numerical response. Line spacing has been normalized. [3]

Statement: It is important for women to have more opportunities in politics.

Here is the most common estimate provided by participants in this study: 71%

Your previous estimate is: [Field-E7_Response]%

Please adjust the slider below if you wish to revise your initial estimate. - Estimated agreement (What % of other study participants do you think would agree with the above statement?)

The example documents the available feedback text. It does not establish the survey's complete display logic or a new mapping from every original export column to the cleaned data. The replication scripts use the existing Statement ID assignments in estimates.csv.

4.2 Attention-check content

The documented checks include an instruction to choose Strongly agree in the agreement section, an explicit reading-carefully response in the estimation section, and an instruction to set a revision slider to 60. These checks are not part of the 12 analyzed gender-norm statements and are not released as analytic outcomes. [2, 3]

5. Pilot reference-value derivation

The reference-value pilot was the earlier survey described in Chen's May 2025 thesis, Peer Support for Inequitable Gender Norms: A Mixed-Methods Study of Perception and Learning. It collected personal agreement and estimates for 18 statements, together with demographic, network, and learning-preference measures. The retained sample contained 879 U.S. adults aged 18–64. The present release includes only the personal-agreement responses for the 12 items used in the main experiment. The thesis's agent-based model and its other survey measures are outside this release. [4]

Items were matched by statement wording and assigned the main-study release IDs. The original pilot item-to-release mapping is shown below. The agreement text is retained unchanged; the numeric values use the same 1–5 coding as estimates.csv.

Table 5. Pilot items and reference calculation.

Release ID	Pilot item	Agree/strongly agree	Agreement (%)	Reference (%)
1	G3	818 / 879	93.0603	93
2	G5	842 / 879	95.7907	96
3	G7	796 / 879	90.5575	91
4	G15	530 / 879	60.2958	60
5	G17	625 / 879	71.1035	71
6	G9	831 / 879	94.5392	95
7	G4	86 / 879	9.7838	10
8	G6	40 / 879	4.5506	5
9	G14	25 / 879	2.8441	3
10	G16	129 / 879	14.6758	15
11	G18	130 / 879	14.7895	15
12	G10	67 / 879	7.6223	8

For each statement, count Agree and Strongly agree responses, divide by the number of valid responses, multiply by 100, and round to the nearest whole percentage point. Each released statement has 879 valid responses. There are no half-percentage-point rounding ties. Do not reverse-code equitable statements for this calculation. The pilot-reference-values.py script checks all 12 calculated integers against statements.csv and writes a separate verification table without replacing either input.

6. Qualitative coding procedure

The released workbook contains two coding stages. The first 181 responses were used for codebook development; their two independent coder assignments are retained in `pilot_coding`. The remaining 722 responses are in `final_coding`, together with the independent assignments and final consensus codes. For the present sharing plan, frequencies and percentages use only `final_coding`. No consensus assignments are inferred for the codebook-development subset. [2, 5]

The unit of analysis is one response. The final coding assigns a single theme. When a response includes multiple rationales, the documented hierarchy prioritizes revision-related over non-revision rationales, then condition-specific over general themes. Where only multiple general rationales remain, the most salient mechanism is selected from its emphasis and elaboration. Disagreements in the final coding were resolved through discussion. [2]

For a condition-specific percentage, divide the number of `final_coding` records with the target consensus code by the total `final_coding` records in that condition, then multiply by 100. The condition denominators are Age 143, Conformity 147, Individual 142, Prestige 146, and Season 144. Include records with Revised = 0, because non-revision explanations are also coded. [5]

Inter-rater agreement uses the two independent coder columns within each coding stage, not the consensus column. Use unweighted Cohen's kappa for nominal categories. The calculation is described in `DATA_DICTIONARY.md`. No qualitative-analysis script is distributed. [2, 5]

6.1 General themes

The following definitions reproduce Supplementary Table S5, with labels and spacing normalized. Examples remain in the supplementary codebook. [2]

1. Normative Alignment

Revision motivated by a desire to reduce discrepancy between one's estimate and the perceived group norm, or by the belief that the group's estimate reflects a common, accurate, or representative view. This theme captures sensitivity to social consensus information and includes conformity-driven adjustment as well as anchoring toward the displayed reference value.

2. Discrepancy Magnitude Sensitivity

Revision or non-revision driven by the perceived size of the gap between one's initial estimate and the reference value. Responses explicitly referencing the estimate as "far off," "way off," or "very off" were coded as revision based on large discrepancy, whereas references to being "close" or "close enough" were coded as non-revision based on small discrepancy. Responses describing the estimate as merely "off," "a little off," or "slightly off" without clear magnitude qualification were not assigned to this category and were evaluated under other themes.

3. Confidence-Weighted Updating

Revision or non-revision driven by the participant's subjective confidence in their initial estimate. Low confidence (e.g., "not sure," "uncertain") justifies revision, whereas high confidence justifies maintaining the original estimate despite exposure to social information. Explicit statements of confidence (e.g., "confident," "trust my gut," "good instinct," "believe in my reasoning") were included in this category.

4. Reflective Reconsideration

Revision following deliberate reassessment of one's prior reasoning. Participants describe actively rethinking, recalculating, or reevaluating their estimate after exposure to the reference value, or engaging in general reconsideration even in the absence of condition-specific information.

5. Prior Overgeneralized Pessimism

Revision justified by recognition that one's initial estimate reflected an overly negative assumption about the broader population. Participants explicitly acknowledge that their prior belief overstated pessimism relative to the reference value.

6. Population Scope Adjustment

Reasoning that explicitly addresses how the displayed subgroup (e.g., 40+, summer-likers, individuals without graduate degrees) relates to the broader population. Participants adjust their estimate by integrating, extrapolating, or comparing subgroup proportions, or by considering the demographic scope or representativeness of the reference group.

7. Other

Responses that do not map onto the above mechanisms or reflect highly individualized reasoning patterns without thematic recurrence.

6.2 Condition-specific themes

Generational Ideology Assumption (A1)

Participants interpret the 40+ group as holding systematically different political or social attitudes (e.g., more conservative or more liberal). Revision decisions are explicitly grounded in beliefs about generational ideological orientation rather than solely in the numerical estimate. References to younger generations (e.g., "under 40 are more liberal/progressive") were coded in this category when such comparisons implicitly characterized the 40+ group's orientation. When generational ideology was explicitly invoked, responses were coded as A1 rather than Population Scope Adjustment.

Age-Based Knowledge Attribution (A2)

Participants treat the 40+ group as more experienced, informed, knowledgeable, or accurate. The displayed estimate is given greater weight because age is used as a proxy for life experience, epistemic competence, or practical understanding.

Majority Accuracy Belief (C1)

Participants explicitly state that the majority, average, or most common response is likely more accurate, representative, or closer to the true population view. Revision reflects trust in aggregation as a mechanism for reducing individual error. This category is restricted to the conformity condition and requires explicit reference to the majority or average as epistemically reliable. In contrast to Normative Alignment, which captures general sensitivity to group norms, C1 requires explicit endorsement of the superiority of the majority.

Majority Distrust (C2)

Participants discount the group average because they believe crowds can be biased, uninformed, or influenced by herd behavior. Non-revision reflects skepticism toward collective judgment.

Absence of External Cue (I1)

Participants report no revision because no additional reference information was presented. Stability of estimates is attributed to lack of new input rather than evaluation of a social source.

Elite–Mass Disconnect (E1)

Resistance to revision based on the belief that highly educated people are ideologically biased, socially detached, or unrepresentative of “ordinary people.”

Elite Deference (E2)

Revision driven by the belief that highly educated people possess superior knowledge or insight. Participants treat education as a proxy for epistemic competence.

Cue Irrelevance Assessment (S1)

Participants interpret season preference as unrelated to the issue. Because the cue is seen as non-diagnostic, they discount the reference value.

Random Sample Inference (S2)

Participants reason that a randomly defined subgroup (e.g., summer-likers) likely approximates the broader population distribution. The cue is interpreted as statistically representative despite being arbitrary.

7. Sources and scope

[1] Main_text REVISED.docx, Sections 2.3–2.5 and the intervention description. These sections supply the documented target population, main procedure, source conditions, and outcome definitions.

[2] Supplementary_information.pdf, Sections 1D–1J, Figure S1, and Tables S1/S5. These supply the statement bank, source framing, procedure diagram, attention-check descriptions, and coding definitions. Numerical qualitative summaries are not copied from the older supplementary tables.

[3] experiment_09.02.csv, question-label row. Only nonparticipant question text is reproduced here. Demographic prompts, confidence and open-response wording, auxiliary items, and the available feedback example come from this row. The original response export is not included in the public cleaned-data package.

[4] Chen, Z. (2025). Peer Support for Inequitable Gender Norms: A Mixed-Methods Study of Perception and Learning. University of Chicago MA thesis, Data and Methods sections; original pilot survey question labels and the supplied preprocessing notebook. The release uses the retained 879-person sample and verifies the twelve reference calculations.

[5] qualitative_coding_final(20261007-024511).xlsx, released as data/qualitative_coding_final.xlsx. Its worksheet membership and saved codes determine the frequency-analysis denominators described here.

The companion provides the available text and documented procedures relevant to the cleaned-data package. It does not reproduce a complete original consent document, every unused survey option, browser-level behavior, or executable randomization/display logic. It should not be presented as the original fielded Qualtrics project. The final supplementary document remains the separate source for the manuscript’s complete methods and results.

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